

FANUC Robot series

R-30iB and R-30iB Plus CONTROLLER

SKS Weld Equipment Library Option

SETUP AND OPERATIONS MANUAL

MAROBKSW02151E REV. C

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FANUC conducts courses on its systems and products on a regularly scheduled basis at the company's world headquarters in Rochester Hills, Michigan. For additional information contact

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One or more of the following U.S. patents might be related to the FANUC products described in this manual.

FANUC America Corporation Patent List

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Conventions

WARNING

Information appearing under the "WARNING" caption concerns the protection of personnel. It is boxed and bolded to set it apart from the surrounding text.

CAUTION

Information appearing under the "CAUTION" caption concerns the protection of equipment, software, and data. It is boxed and bolded to set it apart from the surrounding text.

Note Information appearing next to NOTE concerns related information or useful hints.

Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can be best determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system—including the design, implementation, and debugging of application programs—must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and not replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.

- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points*. Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.

- Before teaching, visually inspect the robot and work envelope to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

WARNING

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

- Using a low motion speed, single step the program for at least one full cycle.
- Using a low motion speed, test run the program continuously for at least one full cycle.
- Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.

- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue its activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

 WARNING

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.
- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

 WARNING

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

NOTE: Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

ADDITIONAL SAFETY CONSIDERATIONS FOR PAINT ROBOT INSTALLATIONS

Process technicians are sometimes required to enter the paint booth, for example, during daily or routine calibration or while teaching new paths to a robot. Maintenance personnel also must work inside the paint booth periodically.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment usually is provided by the paint shop supervisor.

Although paint booth hazards have been minimized, potential dangers still exist. Therefore, today's highly automated paint booth requires that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between the vehicle moving along the conveyor and the robot(s), hood/deck and door opening devices, and high-voltage electrostatic tools.



CAUTION

Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic and exercise operation
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is initiated at the System Operator Console (SOC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth or in a designated safe area within the booth whenever automatic mode is initiated at the SOC or MCP.

In automatic mode, the robots will execute the path movements they were taught during teach mode, but generally at production speeds.

When process and maintenance personnel run diagnostic routines that require them to remain in the paint booth, they must stay in a designated safe area.

Paint System Safety Features

Process technicians and maintenance personnel must become totally familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features:

- Most paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.

- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.
- EMERGENCY STOP buttons are located on the robot controller and teach pendant. Become familiar with the locations of all E–STOP buttons.
- An intrinsically safe teach pendant is used when teaching in hazardous paint atmospheres.
- A DEADMAN switch is located on each teach pendant. When this switch is held in, and the teach pendant is on, power is applied to the robot servo system. If the engaged DEADMAN switch is released or pressed harder during robot operation, power is removed from the servo system, all axis brakes are applied, and the robot comes to an EMERGENCY STOP. Safety interlocks within the system might also E–STOP other robots.

 WARNING

An EMERGENCY STOP will occur if the DEADMAN switch is released on a bypassed robot.

- Overtravel by robot axes is prevented by software limits. All of the major and minor axes are governed by software limits. DCS (Dual Check Safety), limit switches and hardstops also limit travel by the major axes.
- EMERGENCY STOP limit switches and photoelectric eyes might be part of your system. Limit switches, located on the entrance/exit doors of each booth, will EMERGENCY STOP all equipment in the booth if a door is opened while the system is operating in automatic or manual mode. For some systems, signals to these switches are inactive when the switch on the SOC is in teach mode.
- When present, photoelectric eyes are sometimes used to monitor unauthorized intrusion through the entrance/exit silhouette openings.
- System status is monitored by computer. Severe conditions result in automatic system shutdown.

Staying Safe While Operating the Paint Robot

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

 WARNING

Observe all safety rules and guidelines to avoid injury.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

 WARNING

Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.

- Know the work area of the entire paint station (workcell).
- Know the work envelope of the robot and hood/deck and door opening devices.
- Be aware of overlapping work envelopes of adjacent robots.
- Know where all red, mushroom-shaped EMERGENCY STOP buttons are located.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Make sure that the work area near the robot is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the robot will perform BEFORE starting automatic mode.
- Make sure all personnel are outside the paint booth before you turn on power to the robot servo system.
- Never enter the work envelope or paint booth before you turn off power to the robot servo system.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Stay out of areas where you might get trapped between a moving robot, conveyor, or opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of guns or bells.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.

Special Precautions for Combustible Dusts (Powder Paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust can not enter the robot.
- A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust. Do not energize the robot until you have performed the following steps.
 1. Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.
 2. When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
 3. Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
 4. When the dust has been removed, the covers must be replaced immediately.
 5. Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.



WARNING

When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials, and NFPA 70 National Electrical Code.

- **Grounding:** All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.
- **High Voltage:** High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.
- Avoid any accumulation of combustible vapors or coating matter.
- Follow all manufacturer recommended cleaning procedures.
- Make sure all interlocks are operational.

- No smoking.
- Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
- Disable all air and paint pressure to bell.
- Verify that the lines are not under pressure.

Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

- Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
- When a maintenance technician is repairing or adjusting a robot, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the EMERGENCY STOP button. This person must understand the robot and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Always return the robot to the “home” position before you disarm it.
- Never use machine power to aid in removing any component from the robot.
- During robot operations, be aware of the robot’s movements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before you clean the robot.
- When using vinyl resin observe the following:
 - Wear eye protection and protective gloves during application and removal.
 - Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
 - If there is contact with the skin, wash with water.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.
- When using paint remover observe the following:
 - Eye protection, protective rubber gloves, boots, and apron are required during booth cleaning.
 - Adequate ventilation is required. Overexposure could cause drowsiness.
 - If there is contact with the skin or eyes, rinse with water for at least 15 minutes. Then seek medical attention as soon as possible.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.

Table of Contents

1	OVERVIEW	1-1
2	ETHERNET	2-1
2.1	PRELIMINARY STEPS	2-1
2.1.1	Clear Unused Ethernet I/O Assignments	2-1
2.1.2	Configure Multi-Process and Multi-Equipment	2-3
2.1.3	Plan the I/O range	2-3
2.1.4	Configure the SKS IP Address	2-4
2.2	WELD POWER SUPPLY SELECTION, SETUP, AND VERIFICATION	2-5
2.2.1	Weld Power Supply Selection	2-5
2.2.2	Verify Scanner Operation	2-6
2.2.3	Verify the Weld I/O Assignments.....	2-6
2.2.4	Verify the Digital I/O Assignments	2-7
2.2.5	Verify the Group I/O Assignments	2-10
2.2.6	System Variable Settings	2-11
2.3	WELD PROCEDURE DATA SCREEN.....	2-12
2.4	TOUCH SENSING.....	2-13
2.5	SYNCHROWELD™	2-14
2.5.1	TCP Speed Output Setup	2-14
2.5.2	Weld Speed Output Setup.....	2-15

1 Overview

1 OVERVIEW

The SKS Weld Equipment Library Option (R864) allows you to configure ArcTool to communicate with SKS arc welding power supplies via an Ethernet IP connection to the SKS FB5 Fieldbus Interface.

The Ethernet/IP Scanner option R785 is required to use this option.

The SKS Weld Equipment Library is supported by ArcTool beginning with version V8.30P06.

The SKS Weld Equipment Library only supports the Weld Procedure programming technique with Weld Start and Weld End instructions. This is the default method of weld programming ArcTool.

Note that Multi-pass programming in the TP program header is not supported with the Weld Procedures programming method.

The SKS FB5 Fieldbus Interface supports other interfaces such as Devicenet and Profibus. The SKS Weld Equipment Library Option only supports the Ethernet interface.

The SKS UNI5A Universal Robot Interface can be used to interface to a FANUC robot; however, the SKS Weld Equipment Library does not provide a means to setup that configuration.

2 ETHERNET

2 ETHERNET

2.1 Preliminary Steps

Before using the ArcTool setup menu at controlled start to select the SKS ENET interface you should perform these preliminary steps:

1. Clear Unused Ethernet I/O Assignments
2. Configure Multi-Process and Multi-Equipment
3. Plan the range of I/O port numbers used by the welder and other devices in the system.
4. Configure the SKS IP Address

These preliminary steps are described in the next few sections. After these steps are completed the SKS ENET interface can be selected in the controlled start menu as described in Procedure 1-2.

2.1.1 Clear Unused Ethernet I/O Assignments

The SKS Ethernet interface requires the Ethernet/IP Scanner option R785. This option has a default setting that allocates 64 digital I/O signals to the first Ethernet connection as an Adaptor. This digital I/O and the Ethernet connection should be cleared if you do not plan to use it. This should be done before selecting the SKS ENET weld equipment interface. Use Procedure 1-1.

Procedure 1-1 Clearing the Default Ethernet I/O

Step

- 1 Press the MENU key and select I/O. Press TYPE to select Ethernet I/P. You should see a screen similar to the one below.

I/O Ethernet/IP					
Ethernet/IP List(Rack 89)					1/32
Description	TYP	Enable	Status	Slot	
Connection1	ADP	TRUE	OFFLINE	1	
Connection2	ADP	FALSE	OFFLINE	2	
[TYPE] PING CONFIG HELP					

- 2 Cursor to line 1, column 3. Change Connection1 Enable state from TRUE to FALSE.

- 3** Cursor to line 1, column 1. Press F4, CONFIG. The screen should be similar to the one shown below.

```

I/O Ethernet/IP
Adapter configuration :                1/3

      Description :      Connection1
      Input size (words) : 4
      Output size(words) : 4
      Alarm Severity : WARN

      Scanner IP : *****
      API 0=>T :      0
      API T=>0 :      0

[ TYPE ]          PREV          HELP

```

- 4** Change the Input size and the Output size to 0 and then press F3, PREV.
- 5** Press the F1, TYPE and select Digital I/O. Press F2, CONFIG. The screen should be similar to the one shown below. The 64 signals associated with Rack 89 Slot 1 correspond to the default Connection1.

```

I/O Digital Out
                                           1/2
#      RANGE      RACK SLOT START STAT.
1 DO[   1- 64]    89   1     1 ACTIV
2 DO[  65- 512]    0   0     0 UNASG

[ TYPE ] MONITOR IN/OUT  DELETE  HELP

```

- 6** Clear the digital output assignments for Rack 89 Slot 1 by moving the cursor to this line and pressing the DELETE function key.
- 7** Press F3, IN/OUT to select the digital input screen.
- 8** Clear the digital input assignments for Rack 89 Slot 1 by moving the cursor to this line and pressing the DELETE function key.
- 9** Cycle power on the robot controller.

2.1.2 Configure Multi-Process and Multi-Equipment

The SKS ENET interface does not require the ArcTool multi-process feature. It will work with it enabled or with it disabled. If you have other weld equipment connected to this robot that do require the multi-process feature, do not disable multi-process.

If you are connecting multiple weld equipment to the robot controller the multi-equipment option must be installed. You should set the correct number of equipment to be used in the controlled start setup menu and cycle power before selecting the SKS ENET interface.

Disabling multi-process and setting the number of weld equipment (and cycling power) can be done at the same time in the controlled start setup menu or they can be done as separate steps. These menu items are located near the bottom of the menu. From line 1 of the ArcTool controlled start setup menu you can cursor up to display these items.

```
ArcTool Setup

  8 Multi-process:           DISABLED
  9 Number of weld schedules:    32
 10 Number of weld equipment:    1

[ TYPE ]                     HELP
```

2.1.3 Plan the I/O range

If your robot system has other I/O devices it may be important to you where the welding I/O signals are located within the overall I/O map.

The SKS ENET interface uses 96 digital inputs and 96 digital outputs.

During the controlled start setup procedure you have the opportunity to enter a start point for the Digital I/O signals and for the Group I/O signals. You can also allow the signal numbering to begin at the first available area.

Start points begin one after a multiple of 8 (1, 9, 17, etc.) The number you enter will be used to search for an available I/O range. The full range of contiguous signals (96 I/O) must be available without conflict with a previous assignment.

2.1.4 Configure the SKS IP Address

The SKS FB5 interface is designed to use the IP address: 192.168.0.X, where X = a binary switch value. The dip switch is located on the FB5.

In a later step, when the SKS ENet setup is selected at controlled start, Ethernet port 2 of the robot will be given the IP address 192.168.0.0. The Ethernet connection for the SKS will be assigned too as shown below. The number depends on the equipment number. If you only have one welder, it is equipment 1.

192.168.0.0 (Robot Port 2)
192.168.0.1 (Equipment 1)
192.168.0.2 (Equipment 2)
192.168.0.3 (Equipment 3)
192.168.0.4 (Equipment 4)



This picture shows the correct switch settings for 192.168.0.1 (Equipment 1). The LSB is ON to yield a binary one.

2.2 Weld Power Supply Selection, Setup, and Verification

After completing the preliminary steps of the previous section, the weld power supply setup begins by selecting the SKS ENET interface in the controlled start setup menu.

2.2.1 Weld Power Supply Selection

Use the following procedure to configure ArcTool for operation with a SKS weld power supply.

Procedure 1-2 Selecting a SKS Weld Power Supply at Controlled Start Conditions

- The Ethernet/IP Scanner option R785 is installed.
- The SKS Weld Equipment Library Option R864 is installed.
- You have completed the Preliminary Steps in section 1.2.
- The ArcTool Setup menu is displayed at the Controlled Start as shown below.

ArcTool Setup		6/10
1 F Number:	F00000	
2 Welding setup:	USA	
3 Wire speed units:	IPM	
4 Weld speed units:	IPM	
5 Weld speed:	50	
6 Manufacturer:	General Purpose	
7 Model:	MIG (Volts, WFS)	
Press FCTN then START (COLD) when done.		
[TYPE]	CHECK	[CHOICE] HELP

Step

- 1** Move the cursor to the Manufacturer field and press F4, [CHOICE]. Select SKS. The Model name will default to “SKS with Enet”. This is the only model selection available.
- 2** You will be asked if you would like to specify a starting digital I/O number. If you choose not to specify a number, the I/O will be located at the first block of adequate size. If you specify a starting number the I/O will begin at or after this point.
- 3** Press FCTN, Select START(Cold), and press ENTER to Cold Start the controller.

2.2.2 Verify Scanner Operation

After power up, visit the Ethernet I/P screen to confirm the robot and weld power supply are communicating.

Press the MENU key and select I/O. Press F1, TYPE and select Ethernet I/P. You should see a screen similar to the one below.

I/O Ethernet/IP				
Ethernet/IP List(Rack 89)				1/32
Description	TYP	Enable	Status	Slot
SKS_EQ1	SCN	TRUE	RUNNING	1
Connection2	ADP	FALSE	OFFLINE	2
Connection3	ADP	FALSE	OFFLINE	3
[TYPE] PING CONFIG HELP				

The scanner status should indicate “RUNNING”.

2.2.3 Verify the Weld I/O Assignments

I/O Weld Out					I/O Weld In				
				1/11					1/10
WELD SIGNAL	TYPE	#	SIM	STATUS	WELD SIGNAL	TYPE	#	SIM	STATUS
1 [File	] GO[	2]	U	0.0	1 [Voltage	] GI[	1]	U	0.0
2 [Program	] GO[	3]	U	0.0	2 [Current	] GI[	2]	U	0.0
3 [Weld start	] DO[	1]	U	OFF	3 [Wire feed	] GI[	3]	U	0.0
4 [Gas start	] DO[	2]	U	OFF	4 [Arc detect	] DI[	3]	U	OFF
5 [Inch forward	] DO[	3]	U	OFF	5 [Power fault	] DI[	5]	U	OFF
6 [Inch backward	] DO[	4]	U	OFF	6 [Wirestick	] DI[	6]	U	OFF
7 [Proc select 1	] DO[	14]	U	OFF	7 [Gas fault	] [****]	*	***	
8 [Proc select 2	] DO[	15]	U	OFF	8 [Wire fault	] [****]	*	***	
9 [Wire stick cmd	] DO[	6]	U	OFF	9 [Water fault	] [****]	*	***	
10 [Wire stick alarm]	[****]	*	***		10 [Arc enable	] [****]	*	***	
11 [Proc select 3	] [****]	*	***		[TYPE] HELP IN/OUT SIMULATE UNSIM >				
[TYPE] HELP IN/OUT SIMULATE UNSIM >									

Pressing the SHIFT-Wire +/- hard keys feeds the filler wire. This can be used to help verify the digital output operation.

Pressing the SHIFT Gas hard key will purge the gas. The Gas start Output signal will be turned on for a short time.

2.2.4 Verify the Digital I/O Assignments

The digital I/O signals were automatically configured during the controlled start setup. Comments were added to all the I/O signals.

I/O Digital Out					I/O Digital In				
	#	SIM	STATUS	1/512		#	SIM	STATUS	1/512
DO[	1]	U	OFF	[E1 Weld start]	DI[	1]	U	OFF	[E1 Ready]
DO[	2]	U	OFF	[E1 Gas start]	DI[	2]	U	OFF	[E1 Proc active]
DO[	3]	U	OFF	[E1 Inch fwd]	DI[	3]	U	OFF	[E1 Arc detect]
DO[	4]	U	OFF	[E1 Inch bwd]	DI[	4]	U	OFF	[E1 Touch fdbk]
DO[	5]	U	OFF	[E1 Last seam]	DI[	5]	U	OFF	[E1 Alarm]
DO[	6]	U	OFF	[E1 Touch cmd]	DI[	6]	U	OFF	[E1 Wirestuck]
DO[	7]	U	OFF	[E1 Reserved]	DI[	7]	U	OFF	[E1 DIN 0]
DO[	8]	U	OFF	[E1 Reserved]	DI[	8]	U	OFF	[E1 Comm OK]
DO[	9]	U	OFF	[E1 File 1]	DI[	9]	U	OFF	[E1 Alarm 1]
DO[	10]	U	OFF	[E1 File 2]	DI[	10]	U	OFF	[E1 Alarm 2]
DO[	11]	U	OFF	[E1 File 3]	DI[	11]	U	OFF	[E1 Alarm 3]
DO[	12]	U	OFF	[E1 File 4]	DI[	12]	U	OFF	[E1 Alarm 4]
DO[	13]	U	OFF	[E1 File 5]	DI[	13]	U	OFF	[E1 Alarm 5]
DO[	14]	U	OFF	[E1 Group 1]	DI[	14]	U	OFF	[E1 Alarm 6]
DO[	15]	U	OFF	[E1 Group 2]	DI[	15]	U	OFF	[E1 Alarm 7]
DO[	16]	U	OFF	[E1 Reserved]	DI[	16]	U	OFF	[E1 Auto key]
DO[	17]	U	OFF	[E1 Program 1]	DI[	17]	U	OFF	[E1 Synchroweld]
DO[	18]	U	OFF	[E1 Program 2]	DI[	18]	U	OFF	[E1 Reserved]
DO[	19]	U	OFF	[E1 Program 3]	DI[	19]	U	OFF	[E1 Reserved]
DO[	20]	U	OFF	[E1 Reserved]	DI[	20]	U	OFF	[E1 Reserved]
DO[	21]	U	OFF	[E1 Reserved]	DI[	21]	U	OFF	[E1 DIN 1]
DO[	22]	U	OFF	[E1 Reserved]	DI[	22]	U	OFF	[E1 DIN 2]
DO[	23]	U	OFF	[E1 Reserved]	DI[	23]	U	OFF	[E1 DIN 3]
DO[	24]	U	OFF	[E1 Reserved]	DI[	24]	U	OFF	[E1 DIN 4]
DO[	25]	U	OFF	[E1 Reserved]	DI[	25]	U	OFF	[E1 Reserved]
DO[	26]	U	OFF	[E1 Reserved]	DI[	26]	U	OFF	[E1 Reserved]
DO[	27]	U	OFF	[E1 Reserved]	DI[	27]	U	OFF	[E1 Reserved]
DO[	28]	U	OFF	[E1 Reserved]	DI[	28]	U	OFF	[E1 Reserved]
DO[	29]	U	OFF	[E1 Reserved]	DI[	29]	U	OFF	[E1 Reserved]
DO[	30]	U	OFF	[E1 Reserved]	DI[	30]	U	OFF	[E1 Reserved]
DO[	31]	U	OFF	[E1 Reserved]	DI[	31]	U	OFF	[E1 Reserved]
DO[	32]	U	OFF	[E1 Reserved]	DI[	32]	U	OFF	[E1 Reserved]
DO[	33]	U	OFF	[E1 Reserved]	DI[	33]	U	OFF	[E1 Voltage 1]
DO[	34]	U	OFF	[E1 Reserved]	DI[	34]	U	OFF	[E1 Voltage 2]
DO[	35]	U	OFF	[E1 Reserved]	DI[	35]	U	OFF	[E1 Voltage 3]
DO[	36]	U	OFF	[E1 Reserved]	DI[	36]	U	OFF	[E1 Voltage 4]
DO[	37]	U	OFF	[E1 Reserved]	DI[	37]	U	OFF	[E1 Voltage 5]
DO[	38]	U	OFF	[E1 Reserved]	DI[	38]	U	OFF	[E1 Voltage 6]
DO[	39]	U	OFF	[E1 Reserved]	DI[	39]	U	OFF	[E1 Voltage 7]
DO[	40]	U	OFF	[E1 Reserved]	DI[	40]	U	OFF	[E1 Voltage 8]
DO[	41]	U	OFF	[E1 Reserved]	DI[	41]	U	OFF	[E1 Voltage 9]
DO[	42]	U	OFF	[E1 Reserved]	DI[	42]	U	OFF	[E1 Voltage 10]
[TYPE] CONFIG IN/OUT ON OFF >					[TYPE] CONFIG IN/OUT ON OFF >				

I/O Digital Out						I/O Digital In					
#	SIM	STATUS	63/512			#	SIM	STATUS	63/512		
DO[43]	U	OFF	[E1	Reserved	]	DI[43]	U	OFF	[E1	Voltage 11	]
DO[44]	U	OFF	[E1	Reserved	]	DI[44]	U	OFF	[E1	Voltage 12	]
DO[45]	U	OFF	[E1	Reserved	]	DI[45]	U	OFF	[E1	Voltage 13	]
DO[46]	U	OFF	[E1	Reserved	]	DI[46]	U	OFF	[E1	Voltage 14	]
DO[47]	U	OFF	[E1	Reserved	]	DI[47]	U	OFF	[E1	Voltage 15	]
DO[48]	U	OFF	[E1	Reserved	]	DI[48]	U	OFF	[E1	Voltage 16	]
DO[49]	U	OFF	[E1	Reserved	]	DI[49]	U	OFF	[E1	Current 1	]
DO[50]	U	OFF	[E1	Reserved	]	DI[50]	U	OFF	[E1	Current 2	]
DO[51]	U	OFF	[E1	Reserved	]	DI[51]	U	OFF	[E1	Current 3	]
DO[52]	U	OFF	[E1	Reserved	]	DI[52]	U	OFF	[E1	Current 4	]
DO[53]	U	OFF	[E1	Reserved	]	DI[53]	U	OFF	[E1	Current 5	]
DO[54]	U	OFF	[E1	Reserved	]	DI[54]	U	OFF	[E1	Current 6	]
DO[55]	U	OFF	[E1	Reserved	]	DI[55]	U	OFF	[E1	Current 7	]
DO[56]	U	OFF	[E1	Reserved	]	DI[56]	U	OFF	[E1	Current 8	]
DO[57]	U	OFF	[E1	Reserved	]	DI[57]	U	OFF	[E1	Current 9	]
DO[58]	U	OFF	[E1	Reserved	]	DI[58]	U	OFF	[E1	Current 10	]
DO[59]	U	OFF	[E1	Reserved	]	DI[59]	U	OFF	[E1	Current 11	]
DO[60]	U	OFF	[E1	Reserved	]	DI[60]	U	OFF	[E1	Current 12	]
DO[61]	U	OFF	[E1	Reserved	]	DI[61]	U	OFF	[E1	Current 13	]
DO[62]	U	OFF	[E1	Reserved	]	DI[62]	U	OFF	[E1	Current 14	]
DO[63]	U	OFF	[E1	Reserved	]	DI[63]	U	OFF	[E1	Current 15	]
DO[64]	U	OFF	[E1	Reserved	]	DI[64]	U	OFF	[E1	Current 16	]
DO[65]	U	OFF	[E1	Reserved	]	DI[65]	U	OFF	[E1	Wire feed 1	]
DO[66]	U	OFF	[E1	Reserved	]	DI[66]	U	OFF	[E1	Wire feed 2	]
DO[67]	U	OFF	[E1	Reserved	]	DI[67]	U	OFF	[E1	Wire feed 3	]
DO[68]	U	OFF	[E1	Reserved	]	DI[68]	U	OFF	[E1	Wire feed 4	]
DO[69]	U	OFF	[E1	Reserved	]	DI[69]	U	OFF	[E1	Wire feed 5	]
DO[70]	U	OFF	[E1	Reserved	]	DI[70]	U	OFF	[E1	Wire feed 6	]
DO[71]	U	OFF	[E1	Reserved	]	DI[71]	U	OFF	[E1	Wire feed 7	]
DO[72]	U	OFF	[E1	Reserved	]	DI[72]	U	OFF	[E1	Wire feed 8	]
DO[73]	U	OFF	[E1	Reserved	]	DI[73]	U	OFF	[E1	Wire feed 9	]
DO[74]	U	OFF	[E1	Reserved	]	DI[74]	U	OFF	[E1	Wire feed 10	]
DO[75]	U	OFF	[E1	Reserved	]	DI[75]	U	OFF	[E1	Wire feed 11	]
DO[76]	U	OFF	[E1	Reserved	]	DI[76]	U	OFF	[E1	Wire feed 12	]
DO[77]	U	OFF	[E1	Reserved	]	DI[77]	U	OFF	[E1	Wire feed 13	]
DO[78]	U	OFF	[E1	Reserved	]	DI[78]	U	OFF	[E1	Wire feed 14	]
DO[79]	U	OFF	[E1	Reserved	]	DI[79]	U	OFF	[E1	Wire feed 15	]
DO[80]	U	OFF	[E1	Reserved	]	DI[80]	U	OFF	[E1	Wire feed 16	]
DO[81]	U	OFF	[E1	TCP speed 1	]	DI[81]	U	OFF	[E1	TCP speed 1	]
DO[82]	U	OFF	[E1	TCP speed 2	]	DI[82]	U	OFF	[E1	TCP speed 2	]
DO[83]	U	OFF	[E1	TCP speed 3	]	DI[83]	U	OFF	[E1	TCP speed 3	]
DO[84]	U	OFF	[E1	TCP speed 4	]	DI[84]	U	OFF	[E1	TCP speed 4	]
[TYPE] CONFIG IN/OUT ON OFF >						[TYPE] CONFIG IN/OUT ON OFF >					

I/O Digital Out						I/O Digital In					
#	SIM	STATUS	105/512			#	SIM	STATUS	105/512		
DO[85]	U	OFF	[E1 TCP speed 5	]		DI[85]	U	OFF	[E1 TCP speed 5	]	
DO[86]	U	OFF	[E1 TCP speed 6	]		DI[86]	U	OFF	[E1 TCP speed 6	]	
DO[87]	U	OFF	[E1 TCP speed 7	]		DI[87]	U	OFF	[E1 TCP speed 7	]	
DO[88]	U	OFF	[E1 TCP speed 8	]		DI[88]	U	OFF	[E1 TCP speed 8	]	
DO[89]	U	OFF	[E1 Reserved	]		DI[89]	U	OFF	[E1 Reserved	]	
DO[90]	U	OFF	[E1 Reserved	]		DI[90]	U	OFF	[E1 Reserved	]	
DO[91]	U	OFF	[E1 Reserved	]		DI[91]	U	OFF	[E1 Reserved	]	
DO[92]	U	OFF	[E1 Reserved	]		DI[92]	U	OFF	[E1 Reserved	]	
DO[93]	U	OFF	[E1 Reserved	]		DI[93]	U	OFF	[E1 Reserved	]	
DO[94]	U	OFF	[E1 Reserved	]		DI[94]	U	OFF	[E1 Reserved	]	
DO[95]	U	OFF	[E1 Reserved	]		DI[95]	U	OFF	[E1 Reserved	]	
DO[96]	U	OFF	[E1 Reserved	]		DI[96]	U	OFF	[E1 Reserved	]	
DO[97]	*	*	[	]		DI[97]	*	*	[	]	
DO[98]	*	*	[	]		DI[98]	*	*	[	]	
DO[99]	*	*	[	]		DI[99]	*	*	[	]	
DO[100]	*	*	[	]		DI[100]	*	*	[	]	
DO[101]	*	*	[	]		DI[101]	*	*	[	]	
DO[102]	*	*	[	]		DI[102]	*	*	[	]	
DO[103]	*	*	[	]		DI[103]	*	*	[	]	
DO[104]	*	*	[	]		DI[104]	*	*	[	]	
DO[105]	*	*	[	]		DI[105]	*	*	[	]	
[TYPE] CONFIG IN/OUT ON OFF >						[TYPE] CONFIG IN/OUT ON OFF >					

The CONFIG function key can be used to confirm the correct numbers of signals are mapped (96 outputs and 96 inputs).

I/O Digital Out						I/O Digital In					
#	RANGE	RACK	SLOT	START	STAT.	#	RANGE	RACK	SLOT	START	STAT.
1	DO[1- 96]	89	1	1	ACTIV	1	DI[1- 96]	89	1	1	ACTIV
2	DO[97- 512]	0	0	0	UNASG	2	DI[97- 512]	0	0	0	UNASG
Device Name : EthernetIP						Device Name : EthernetIP					
[TYPE] MONITOR IN/OUT DELETE HELP						[TYPE] MONITOR IN/OUT DELETE HELP					

2.2.5 Verify the Group I/O Assignments

The group inputs and outputs signal names should match those shown in the following figure. The port numbers shown below are based on a starting port number of 1.

I/O Group Out					I/O Group In				
#	SIM	VALUE		1/100	#	SIM	VALUE		1/100
G0[1]	U	0	[E1 Group	]	GI[1]	U	0	[E1 Voltage	]
G0[2]	U	0	[E1 File	]	GI[2]	U	0	[E1 Current	]
G0[3]	U	0	[E1 Program	]	GI[3]	U	0	[E1 Wire feed	]
G0[4]	U	0	[E1 TCP speed	]	GI[4]	U	0	[E1 TCP speed	]
G0[5]	*	*	[	]	GI[5]	U	0	[E1 Alarm	]
G0[6]	*	*	[	]	GI[6]	*	*	[	]
G0[7]	*	*	[	]	GI[7]	*	*	[	]
G0[8]	*	*	[	]	GI[8]	*	*	[	]
G0[9]	*	*	[	]	GI[9]	*	*	[	]
G0[10]	*	*	[	]	GI[10]	*	*	[	]
G0[11]	*	*	[	]	GI[11]	*	*	[	]
G0[12]	*	*	[	]	GI[12]	*	*	[	]
G0[13]	*	*	[	]	GI[13]	*	*	[	]
G0[14]	*	*	[	]	GI[14]	*	*	[	]
G0[15]	*	*	[	]	GI[15]	*	*	[	]
G0[16]	*	*	[	]	GI[16]	*	*	[	]
G0[17]	*	*	[	]	GI[17]	*	*	[	]
G0[18]	*	*	[	]	GI[18]	*	*	[	]
G0[19]	*	*	[	]	GI[19]	*	*	[	]
G0[20]	*	*	[	]	GI[20]	*	*	[	]
G0[21]	*	*	[	]	GI[21]	*	*	[	]
Sorted by port number.					Sorted by port number.				
[TYPE] CONFIG IN/OUT SIMULATE UNSIM >					[TYPE] CONFIG IN/OUT SIMULATE UNSIM >				

The CONFIG function key can be used to show how many signals (NUM PTS) are mapped to each port.

I/O Group Out					I/O Group In				
G0 #	RACK	SLOT	START PT	NUM PTS	GI #	RACK	SLOT	START PT	NUM PTS
1	89	1	14	2	1	89	1	33	16
2	89	1	9	5	2	89	1	49	16
3	89	1	17	3	3	89	1	65	16
4	89	1	81	8	4	89	1	81	8
5	0	0	0	0	5	89	1	9	7
					6	0	0	0	0
[TYPE] MONITOR IN/OUT					[TYPE] MONITOR IN/OUT				
HELP >					HELP >				

2.2.6 System Variable Settings

Choosing the SKS Ethernet interface during controlled start disables burnback control by the robot and enables 2 Weld End features. This allows burnback to be controlled by the SKS weld power supply and requires the robot to wait for the machine to complete welding.

\$awspcr.\$bback_enabl = FALSE The burnback weld schedule values and time delay are not used during the Weld End sequence.

\$awepr[eq].\$ae_hold_ao = TRUE The weld command signals (analog or group outputs) are left at their last values when the Weld End completes.

\$awepr[eq].\$ae_chk_di = TRUE If \$ae_chk_di is TRUE the execution of an Weld End will delay after burnback until the Process Active input signal is OFF before proceeding with postflow and wirestick checking. The maximum delay before an error is reported is controlled by \$awepr[eq].\$max_end_time

\$aweioex[eq].\$proc_active.\$port_num This port number is assigned to the Process Active signal of the SKS interface. The Weld End will wait for this signal to be OFF. If the time exceeds the max_end_time the error "ARC-250 Process active ON after weld" is reported.

2.3 Weld Procedure Data Screen

When you create a new weld procedure it will have the default values shown below.

DATA Weld Procedure 1					2/6
+	Procedure	1	[		]
	Group	0	[		]
-	Schedules				
Schedule	File	Prog	Speed	Time	
Schedule 1	1.0	2.0	35.0	0.00	
Schedule 2	1.0	2.0	35.0	0.00	
Schedule 3	1.0	2.0	35.0	0.00	
[TYPE] DETAIL [CMND][VIEW] HELP					

The ArcTool features Runin, Burnback, Wirestick Reset, Ramping, and OnTheFly are not displayed in the weld procedure because they are not compatible with the SKS interface. The interface does not support direct control of the welding parameters.

The welding voltage and wire feed speed can be set on the SKS Q80 with the touch screen user interface. The values are stored in a **program**. The programs are associated with **files** and the files are in **groups**. The process select number in an ArcTool Weld Procedure is used to select a group on the Q80. Each weld schedule in the weld procedure has parameters for specifying the file and program numbers.

A process select number is defined in each weld procedure. It is labeled **Group** for the SKS machine. It has a range of 0 to 3. When a weld procedure is selected it sets a group output to select the SKS group. The system variable \$awepcr[eq].\$cmnd1offset should be 0 so there is no offset between the group number displayed in the DATA screen and the GO command value.

The file and program group outputs are set when a Weld Start is executed. File has a range of 1 to 31. Program has a range of 2 to 7.

Note the SKS does not permit changing the Group or the File number while welding. The program number can be changed while welding with a weld schedule change.

The robot controls the gas flow during Weld Starts. The gas purge and gas preflow times can be set in each Weld Procedure. The preflow setting available on the weld controller is overridden by the robot.

2.4 Touch Sensing

If the touch sense option is installed on the robot controller the touch sense digital input and output signals must be assigned in the Touch Sense I/O Setup menu. The port numbers can be obtained from the digital input and output screens as shown below. This example is for a start point of 1. Your signal numbers may differ.

Touch I/O Setup 2/4

NAME	VALUE
1 Sensor port type:	DI
2 Sensor port number:	4
3 Circuit port type:	DO
4 Circuit port number:	6

[TYPE] HELP

I/O Digital Out 1/512

#	SIM	STATUS	
DO[1]	U	OFF	[E1 Weld start]
DO[2]	U	OFF	[E1 Gas start]
DO[3]	U	OFF	[E1 Inch fwd]
DO[4]	U	OFF	[E1 Inch bwd]
DO[5]	U	OFF	[E1 Last seam]
DO[6]	U	OFF	[E1 Touch cmd]
DO[7]	U	OFF	[E1 Reserved]
DO[8]	U	OFF	[E1 Reserved]
DO[9]	U	OFF	[E1 File 1]
DO[10]	U	OFF	[E1 File 2]
DO[11]	U	OFF	[E1 File 3]
DO[12]	U	OFF	[E1 File 4]
DO[13]	U	OFF	[E1 File 5]
DO[14]	U	OFF	[E1 Group 1]
DO[15]	U	OFF	[E1 Group 2]
DO[16]	U	OFF	[E1 Reserved]
DO[17]	U	OFF	[E1 Program 1]
DO[18]	U	OFF	[E1 Program 2]
DO[19]	U	OFF	[E1 Program 3]

Sorted by port number.
[TYPE] CONFIG IN/OUT ON OFF >

I/O Digital In 1/512

#	SIM	STATUS	
DI[1]	U	OFF	[E1 Ready]
DI[2]	U	OFF	[E1 Proc active]
DI[3]	U	OFF	[E1 Arc detect]
DI[4]	U	OFF	[E1 Touch fdbk]
DI[5]	U	OFF	[E1 Alarm]
DI[6]	U	OFF	[E1 Wirestuck]
DI[7]	U	OFF	[E1 DIN 0]
DI[8]	U	OFF	[E1 Comm OK]
DI[9]	U	OFF	[E1 Alarm 1]
DI[10]	U	OFF	[E1 Alarm 2]
DI[11]	U	OFF	[E1 Alarm 3]
DI[12]	U	OFF	[E1 Alarm 4]
DI[13]	U	OFF	[E1 Alarm 5]
DI[14]	U	OFF	[E1 Alarm 6]
DI[15]	U	OFF	[E1 Alarm 7]
DI[16]	U	OFF	[E1 Auto key]
DI[17]	U	OFF	[E1 Synchroweld]
DI[18]	U	OFF	[E1 Reserved]
DI[19]	U	OFF	[E1 Reserved]

Sorted by port number.
[TYPE] CONFIG IN/OUT ON OFF >

2.5 Synchronweld™

SKS weld equipment supports a feature known as “Synchronweld™” that correlates the robot velocity while welding with the wire feed speed and power supply output to maintain a fixed weld metal deposition rate. The robot speed must be sent to the SKS as a group output signal. This can be done by either of these two options:

- TCP Speed Output (supports one robot)
- Weld Speed Output J682 (supports multiple robots)

Please refer to the SKS user manual to enable and setup the Synchronweld option in the Q80 user interface. Note that the Robot Speed value and the Wire Feed Speed reference establish the desired deposition rate ratio that will be maintained when the actual robot speed is less than the specified speed. This applies to programmed speed changes as well as deceleration and acceleration speed changes that typically occur at weld start/end and changes in path direction. The minimum Synchronweld wire feed speed should also be specified in the Q80 user interface.

2.5.1 TCP Speed Output Setup

The robot software option TCP Speed Output J694 can be configured to set the TCP Speed group output in the SKS interface (for only one robot group). This is GO[4] in the example in section 2.2.5. You can find more details about the setup of the TCP Speed Output option in the ArcTool Setup & Operations Manual. For the SKS interface, the detailed setup menu is shown in the following figure:

TCP speed output	0
TCP_SPD[1]	
1/6	
1 Enable:	ON
2 Target:	GO [4]
3 Min. value:	0.000
4 Max. value:	900.000
5 Min. speed (mm/sec):	0.000
6 Max. speed (mm/sec):	300.000

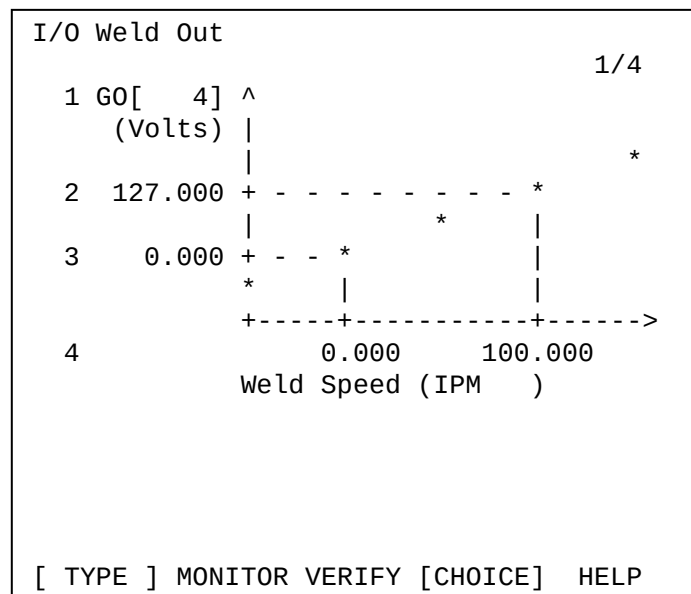
These min and max values are used to scale the output signal and thereby change the units from mm/sec to 0.02m/min.

2.5.2 Weld Speed Output Setup

The robot software option *Weld Speed Output J682* can be configured to set the TCP Speed group output in the SKS interface (for one or more robot groups). This is GO[4] in the example in section 2.2.5. You can find more details about the setup of the Weld Speed Output option in the ArcTool Setup & Operations Manual. Setting the system variable \$awepcr[1].\$enb_wld_spd = TRUE enables this feature and adds the Weld Speed signal to the Weld IO Output menu shown below.

I/O Weld Out					1/12
	WELD SIGNAL	TYPE #	SIM	STATUS	
1	[File	] GO[2]	U	0.0	
2	[Program	] GO[3]	U	0.0	
3	[Weld start	] DO[1]	U	OFF	
4	[Gas start	] DO[2]	U	OFF	
5	[Inch forward	] DO[3]	U	OFF	
6	[Inch backward	] DO[4]	U	OFF	
7	[Proc select 1	] DO[14]	U	OFF	
8	[Proc select 2	] DO[15]	U	OFF	
9	[Wire stick cmd	] DO[6]	U	OFF	
10	[Weld Speed	] GO[4]	U	0.0	
11	[Wire stick alarm]	[****]	*	***	
12	[Proc select 3	] [****]	*	***	
[TYPE] HELP IN/OUT SIMULATE UNSIM					

If you cursor to the Weld Speed signal, press NEXT, and then F3, CONFIG, you can set the port type, port number, and the four scaling values as shown below.



This sets values in the system variable \$AWEIOEX[1].\$WELD_SPEED. You may need to also set the values of entries 7, 8, and 9 below. With these settings the value of \$SLOPE is calculated as 1.270. This is the necessary conversion when the weld speed units are IPM.

```

SYSTEM Variables
$AWEIOEX[1].$WELD_SPEED          1/17
 1 $EQ_MIN_IN      0.000
 2 $EQ_MAX_IN      127.000
 3 $EQ_MIN_OUT     0.000
 4 $EQ_MAX_OUT     100.000
 5 $IO_MIN_VAL     0.000
 6 $IO_MAX_VAL     1000.000
 7 $IO_MIN_BIN     0
 8 $IO_ZERO_BIN    0
 9 $IO_MAX_BIN     1000
10 $SLOPE          1.270
11 $INTERCEPT    0.000
12 $PORT_NUM       4
13 $PORT_TYPE      19
14 $WELD_SIGNAL    1
15 $FORMAT         1
16 $NAME           'Weld Speed'
17 $UNITS          'IPM  '

[ TYPE ]

```

If your robot is configured to use other units for weld speed, then the values must be changed. ArcTool supports two other weld speed units: mm/sec and cm/min. The correct values for these units are shown below.

```

SYSTEM Variables
$AWEIOEX[1].$WELD_SPEED          1/17
 1 $EQ_MIN_IN      0.000
 2 $EQ_MAX_IN      300.000
 3 $EQ_MIN_OUT     0.000
 4 $EQ_MAX_OUT     100.000
 5 $IO_MIN_VAL     0.000
 6 $IO_MAX_VAL     1000.000
 7 $IO_MIN_BIN     0
 8 $IO_ZERO_BIN    0
 9 $IO_MAX_BIN     1000
10 $SLOPE          3.000
11 $INTERCEPT    0.000
12 $PORT_NUM       4
13 $PORT_TYPE      19
14 $WELD_SIGNAL    1
15 $FORMAT         1
16 $NAME           'Weld Speed'
17 $UNITS          'mm/sec  '

[ TYPE ]

```

```

SYSTEM Variables
$AWEIOEX[1].$WELD_SPEED          1/17
 1 $EQ_MIN_IN      0.000
 2 $EQ_MAX_IN      50.000
 3 $EQ_MIN_OUT     0.000
 4 $EQ_MAX_OUT     100.000
 5 $IO_MIN_VAL     0.000
 6 $IO_MAX_VAL     1000.000
 7 $IO_MIN_BIN     0
 8 $IO_ZERO_BIN    0
 9 $IO_MAX_BIN     1000
10 $SLOPE          0.500
11 $INTERCEPT    0.000
12 $PORT_NUM       4
13 $PORT_TYPE      19
14 $WELD_SIGNAL    1
15 $FORMAT         1
16 $NAME           'Weld Speed'
17 $UNITS          'cm/min  '

[ TYPE ]

```